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M 006 HORIZONTAL CENTRIFUGAL PUMPS

1 SCOPE OF WORK

The horizontal centrifugal pumps specified in **Part 4, Subsection 1.2 "Summary of Work"**, in **Section 3 Technical Specifications** or as required by the Bidder's design and the associated electrical equipment as detailed in **Clause 8** herein, shall also be subject to the design, testing and quality assurance requirements of this specification as applicable.

2 CODES AND STANDARDS

The Bidder shall-comply with the requirements of **Part 4, Subsection 2.5 "Codes & Standards"**. The Codes and Standards of the following organizations extracted from **Subsection 2.5** or international equivalent standards proposed by Bidder are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials
AWWA	American Water Works Association
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HIS	Hydraulic Institute Standards
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronic Engineers
JIS	Japan Industrial Standard

3 DESIGN AND OPERATING CONDITIONS

A minimum scope of design and operating data shall be submitted by the Bidder for each horizontal centrifugal pump as requested in **Schedule 7 “Technical Data”**.

4 DETAILED REQUIREMENTS

4.1 General

Each pump and its appurtenances and accessories shall conform to the requirements of this Specification.

The Total Head (H) specified by the Bidder to the pump manufacturer shall be calculated based on commercial pipes (not new and clean).

The Bidder shall submit calculation or design criteria for approval for all pumps furnished. Included shall be: Process flow requirement, interface condition at suction source and delivery point; friction losses; margins; NPSH available; selected pump design conditioning reference list for sources of pump sizing criteria.

4.2 Pump Type and Design

The Bidder shall take into consideration pump rpm and suction specific speed for each application. The suction specific speed shall be in accordance with HIS (Hydraulic Institute Standards) recommendation. The final selection of these parameters will be subject to the Owners approval.

Pumps shall be of the horizontally mounted centrifugal type, direct connected to their motor drives. Each pump, except close-coupled types, shall be connected to its motor drive through a flexible coupling.

Each pump and motor drive, except close-coupled types, shall be mounted on a common bedplate.

Pumps shall be of the heavy-duty type suitable for continuous service over the life of the power plant under all operating conditions as specified. Each pump shall operate without undue strain or wear and without damage to any part of the pumping unit. It shall be Bidder's responsibility to determine the exact type

of pump design which will provide such service without operating difficulty or undue replacement of parts.

Pumps shall be designed and constructed for the most severe service conditions under which the pumps will be required to operate. All materials of construction shall be identified with ASTM or AISI standard numbers. All features of design and construction and all materials shall be acceptable to the Owner.

All parts of each pump unit shall be designed to withstand stresses resulting from a full voltage start of the motor.

The Bidder shall be responsible for determining that each pump and driver operate in dynamic balance as a unit without undue vibration.

Wear rings, impellers, shaft sleeves and other parts (either rotating or stationary) forming the pump assembly shall be securely locked in place, liberally designed and renewable.

4.3 Pump Characteristics

Each pump shall have a stable, steadily rising head characteristic curve with decrease in capacity, suitable for the service specified. Each pump and motor combination shall have non overloading characteristics.

Duplicate pumps designed for the same conditions shall operate satisfactorily in parallel with each other or with other pumps of similar characteristics.

The pump characteristics shall include NPSH required (NPSHR) curves. The recommended minimum NPSHR for normal continuous operation shall be indicated on the curves, for all flows including maximum run-out.

A minimum margin of 1 mWC shall be maintained between NPSHR and NPSHA.

4.4 Casings

Pumps shall be of the horizontal or vertical split-case design unless otherwise noted in other specifications, constructed to permit ease of access for inspection and removal of rotating element and replacement of wearing parts without

disconnecting the suction and discharge piping and without disturbing pump alignment.

Lifting lugs or eyes shall be provided as required.

Pump suction and discharge connections shall be flanged unless otherwise specified. Suction and discharge flanges shall match ANSI Standard pipe flanges, as specified.

4.5 Impellers

All rotors shall be dynamically balanced with all rotating parts connected thereto.

4.6 Wearing Rings

Wearing rings shall be provided on all impeller entrances and on both the impeller and casing when so required for severe service. The design of rings shall be such as to be easily renewed and maintained in service and shall reduce leakage to a minimum.

4.7 Shaft and Sleeves

Suitable materials shall be provided for abrasive and or corrosive service.

Shaft sleeves shall be provided on all pump shafts. They shall be designed and fitted to protect the shaft from wear and corrosion at points in contact with stuffing boxes and other wearing parts, and where required, shall also protect the shaft from contact with the liquid being pumped.

4.8 Shaft Seals

Shaft seals shall be manufacturer's standard design suitable for the specified service conditions.

Pumps that operate under partial vacuum on the suction (condensate pumps etc.) shall be provided with a water seal and shall be designed so that the stuffing boxes are under positive pressure from the inside to preclude air leakage under all operating, standby and startup conditions. Provisions shall be made for injection-of water for startup and during standby condition.

Leak-off drains shall be neatly piped to the bedplate drain outlet.

4.9 Bearings and Lubrication

Main bearings shall be of sleeve or ball type.

Thrust Bearings: Amply sized and easily adjustable. Thrust bearings where required shall be "Kingsbury" or "Michell" self-contained type. Double row anti-friction bearings may be provided as an alternate.

Bearings shall be located and arranged to facilitate alignment, maintenance and to permit easy replacement.

Suitable provisions shall be made to prevent throwing of grease or oil.

One end of pump or driver shaft shall be accessible for taking speed readings with a portable tachometer.

4.10 Oil Piping

All external oil piping shall be of all steel construction with welded joints and forged steel fittings, except that steel welding neck or slip-on flanges may be used where required for dismantling and maintenance.

All oil piping shall be thoroughly cleaned before it is installed and all lines shall be thoroughly cleaned out after they are installed so as to preclude the possibility of dirt or any other foreign material being present in the system when the pump(s) are placed in operation.

4.11 Coupling and Guards

A flexible shaft coupling with spacer, of make and type as approved, shall be provided on all pumps, except close-coupled type. The spacer length shall be designed to allow dismantling the pump bearings, shaft seals, coupling halves and pump internals without disturbing the motor or piping. The pump end coupling half shall have a tapered bore.

Coupling guards and all other guards necessary for protection against external moving parts shall be provided on the pumps. Guards shall be of substantial sheet steel construction suitably braced and reinforced and mounted on an angle-iron frame. Guards shall be of a type acceptable to the Owner.

No keys shall extend beyond coupling guards.

4.12 Bedplates

Each pump and driver, with the exception of motor-driven close- coupled pumps, shall be bolted to a common bedplate that shall extend under and beyond all parts of the pump and driver.

Bedplates shall be manufacturer's standard type.

4.13 Shims

The pumping unit design, with exception of close-coupled types, shall provide shim space between the bedplate and motor feet.

4.14 Drains and Vents

All base plates shall be provided with adequate drains.

4.15 Cleaning and Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter. All external unmachined ferrous metal surfaces shall be painted in accordance with the requirement as mentioned in **Part 4 Technical Specification S-420 and S-430**.

5 PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Test

All horizontal centrifugal pumps more than 30kW shall be shop tested in accordance with Part 4 Section 2.6 and **Part 4 Section 3, M-014**.

5.2 Site Test

After equipment is put in service, the Bidder shall run a test on the equipment to ascertain that it is performing properly in every respect. Testing shall be in accordance with the guidelines and requirements contained in the "Hydraulic Institute Standard". Any deficiencies brought out by this test shall be promptly corrected by Bidder.

6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control shall conform to the requirement of **Part 4, subsection 2.7**.

7 INSTRUMENTATION AND CONTROLS

Any special Instrumentation and Control devices integral with the pumps and appurtenances shall be supplied and installed by the Bidder. The Bidder shall include details of any such devices in his Bid Proposal.

Bidder shall furnish a local control panel for emergency, maintenance and testing purposes. The panel shall contain necessary control switches, and indicator lights.

The Bidder shall also provide complete logic and information describing the operation of the Pumps.

Refer to **Part 4 Section 3, I-005** Distributed Digital Control System.

8 ELECTRICAL SPECIFICATIONS

The Bidder shall furnish and install all electrical equipment and accessories required for Horizontal Centrifugal Pumps.

For general electrical requirements, see **Part 4 Section 3, E-001**.

All motors shall be in accordance with **Part 4 Section 3, E-211** for Medium Voltage Motors and **Part 4 Section 3, E-210** for Low Voltage Motors.

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